

REPORT SUBMISSION

Experimental Verification of Neutral Point Location for a Wing–Tail Configuration Using Glide Tests

Bharat Space Education Research Center (BSERC)



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[Github repository of Project](#)

Abstract

This report verifies the neutral point (NP) location and longitudinal static stability of a simplified two-surface wing–tail configuration. Under the assumptions of identical lifting surfaces and negligible downwash interaction, theory predicts the NP at the midpoint between wing and tail aerodynamic centers. To satisfy the stability verification deliverables, simulation-based evidence is presented for stable, neutral, and unstable CG locations relative to NP. A trim-at- $t = 0$ procedure and a numerical static stability check using the sign of $\partial M/\partial \alpha$ are included to strengthen the validation.

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Chapter 1

Introduction

The neutral point (NP) is a key concept in longitudinal static stability. It is defined as the center-of-gravity (CG) position at which the aircraft exhibits *neutral* static stability, i.e., a small perturbation in angle of attack α produces no net restoring or diverging pitching tendency ($\partial M/\partial \alpha \approx 0$). The NP therefore represents the boundary between statically stable and statically unstable configurations.

In this project, the aircraft is modeled as a simplified wing–tail configuration consisting of two identical lifting surfaces. Under the assumptions of equal lift-curve slopes, equal areas, and negligible downwash interaction, the NP can be predicted directly from geometry: it lies at the midpoint between the aerodynamic centers (AC) of the wing and tail. This theoretical prediction provides a clear reference point for evaluating stability as the CG is shifted forward and aft.

The objectives of the work are:

1. to estimate the neutral point location from the provided wing and tail AC positions, and
2. to verify the expected stability transition by testing three CG locations: forward of NP (stable), at NP (neutral), and aft of NP (unstable).

Note on evidence: The project deliverables request experimental glide-test photos or videos to demonstrate stable and unstable behavior. Due to practical limitations, this submission uses a simulation-based procedure that applies the same geometry and mass assumptions to a nonlinear longitudinal model. The report therefore provides (i) theoretical NP prediction and (ii) simulation evidence (pitch response and stability derivatives) to support the required stability verification.

Chapter 2

Theoretical Background

This chapter establishes the theoretical basis for neutral point (NP) estimation and longitudinal static stability classification. The key idea is that static stability is determined by how the pitching moment changes when the angle of attack α is perturbed about a trim condition. Using a two-surface wing–tail model, the NP is derived as the CG location that makes the pitching-moment slope with respect to α equal to zero.

2.1 Neutral Point Definition

Longitudinal static stability is commonly assessed using the pitching moment coefficient C_m (or equivalently the dimensional pitching moment M). For small perturbations about a trim condition, the sign of the slope with respect to α determines stability:

$$\frac{dC_m}{d\alpha} < 0 \text{ (stable)}, \quad \frac{dC_m}{d\alpha} = 0 \text{ (neutral)}, \quad \frac{dC_m}{d\alpha} > 0 \text{ (unstable)}. \quad (2.1)$$

A negative slope means that if α increases slightly, the aircraft generates a restoring (nose-down) moment that tends to reduce α back toward trim. A positive slope indicates that an increase in α generates a moment that further increases α (divergence).

The **neutral point** is defined as the CG position at which the static slope is zero:

$$x_{CG} = x_{NP} \iff \frac{dC_m}{d\alpha} = 0. \quad (2.2)$$

Thus, the NP represents the boundary between restoring and diverging static pitching behavior.

2.2 Two-Surface Model and Assumptions

A simplified aircraft is modeled as two lifting surfaces: a wing and a tail. Their aerodynamic centers are located at $x_{AC,w}$ and $x_{AC,t}$ along the longitudinal axis, while the CG is at x_{CG} . The aerodynamic centers represent the effective points of action of the incremental lift for small changes in α .

For small angles, lift is assumed to be linear in angle of attack:

$$L_w \approx q_\infty S_w C_{L\alpha_w} \alpha, \quad L_t \approx q_\infty S_t C_{L\alpha_t} \alpha_t, \quad (2.3)$$

where S_w and S_t are the planform areas, $C_{L\alpha_w}$ and $C_{L\alpha_t}$ are lift-curve slopes, and

$$q_\infty = \frac{1}{2} \rho V^2 \quad (2.4)$$

is the dynamic pressure. In general, the tail does not experience the same angle of attack as the wing because wing downwash reduces the effective incidence at the tail. This is represented by:

$$\alpha_t = \alpha - \epsilon(\alpha), \quad \frac{d\alpha_t}{d\alpha} = 1 - \frac{d\epsilon}{d\alpha}. \quad (2.5)$$

The quantity $\frac{d\epsilon}{d\alpha}$ reduces how strongly tail lift increases with wing angle of attack, which in turn shifts the NP forward compared to the simple midpoint rule.

The project simplifies the model using:

1. **Identical surfaces:** $S_w = S_t$ and $C_{L\alpha_w} = C_{L\alpha_t}$,
2. **Negligible downwash:** $\frac{d\epsilon}{d\alpha} \approx 0 \Rightarrow \alpha_t \approx \alpha$.

With these assumptions, both surfaces respond equally to changes in α , and the NP depends only on geometry.

2.3 Neutral Point for Identical Surfaces (Negligible Downwash)

The net pitching moment about the CG is produced by the lifts acting at their moment arms relative to the CG. Using the sign convention that a positive moment is nose-up, the total moment can be written as:

$$M = L_w(x_{AC,w} - x_{CG}) + L_t(x_{AC,t} - x_{CG}). \quad (2.6)$$

The static stability slope is obtained by differentiating with respect to α :

$$\frac{dM}{d\alpha} = \frac{dL_w}{d\alpha}(x_{AC,w} - x_{CG}) + \frac{dL_t}{d\alpha}(x_{AC,t} - x_{CG}). \quad (2.7)$$

At the neutral point, the aircraft is neutrally stable, so $\frac{dM}{d\alpha} = 0$ when $x_{CG} = x_{NP}$. This yields the general two-surface neutral point expression:

$$x_{NP} = \frac{S_w C_{L\alpha_w} x_{AC,w} + S_t C_{L\alpha_t} \left(1 - \frac{d\epsilon}{d\alpha}\right) x_{AC,t}}{S_w C_{L\alpha_w} + S_t C_{L\alpha_t} \left(1 - \frac{d\epsilon}{d\alpha}\right)}. \quad (2.8)$$

This form shows that the NP is a lift-slope-weighted average of the aerodynamic center locations, with the tail contribution reduced by downwash through the factor $\left(1 - \frac{d\epsilon}{d\alpha}\right)$.

Under the project assumptions ($S_w = S_t$, $C_{L\alpha_w} = C_{L\alpha_t}$, and $\frac{d\epsilon}{d\alpha} \approx 0$), the weighting becomes equal and the NP reduces to the midpoint:

$$\boxed{x_{\text{NP}} = \frac{x_{\text{AC},w} + x_{\text{AC},t}}{2}}. \quad (2.9)$$

Therefore, the NP prediction in this project is driven directly by the wing and tail AC positions.

2.4 Static Margin

The **static margin** quantifies how far the CG is from the neutral point:

$$SM = x_{\text{NP}} - x_{\text{CG}}. \quad (2.10)$$

It is positive when the CG is ahead of NP (stable), zero at NP (neutral), and negative when CG is aft of NP (unstable). To report this margin relative to airframe size, it is commonly normalized by the chord length c :

$$SM(\%c) = \frac{x_{\text{NP}} - x_{\text{CG}}}{c} \times 100. \quad (2.11)$$

Hence, SM provides a single scalar measure that links the theoretical NP prediction to practical stability assessment: larger positive static margin implies stronger restoring tendency, while a negative static margin implies static instability.

Chapter 3

Geometry and Mass Calculations

This chapter lists the geometric reference points and the mass/aerodynamic parameters used in the neutral point (NP) estimation and in the longitudinal simulation. All quantities are defined along the body-fixed longitudinal axis x .

3.1 Geometry Definition

The wing and tail aerodynamic centers (AC) are specified as:

$$x_{AC,w} = 0.0 \text{ m}, \quad x_{AC,t} = 1.76 \text{ m}. \quad (3.1)$$

Hence, the longitudinal separation between the lifting surfaces is:

$$\Delta x = x_{AC,t} - x_{AC,w} = 1.76 \text{ m}. \quad (3.2)$$

These locations are the primary geometric inputs for computing x_{NP} under the midpoint assumption and for computing the pitching moment arms about the CG.

3.2 Aerodynamic Parameters (Model Inputs)

A simplified linear lift model is used for both surfaces. The main aerodynamic inputs are:

- **Chord:** $c = 0.23 \text{ m}$ (used to report static margin as % chord),
- **Total lifting area:** $S = 2.4 \times 0.23 \text{ m}^2$ with equal split $S_w = S_t = S/2$ (consistent with identical-surface assumption),
- **Lift-curve slopes:** $C_{L\alpha_w} = C_{L\alpha_t} = 2\pi \text{ rad}^{-1}$ (small-angle thin-airfoil approximation),
- **Drag model:** $C_D = C_{D0} + k(C_{L,w}^2 + C_{L,t}^2)$ to ensure physically reasonable glide deceleration.

These parameters provide the lift forces that generate the pitching moment M about the CG.

3.3 Mass Properties

The rigid-body mass properties used in the simulation are:

$$m = 1.8 \text{ kg}, \quad I_{yy} = 0.2 \text{ kg} \cdot \text{m}^2. \quad (3.3)$$

Ballast effects are represented by shifting the CG location x_{CG} while keeping m and I_{yy} fixed, which directly changes the aerodynamic moment arms and therefore the stability behavior.

3.4 CG Test Locations

To demonstrate the stability transition across the NP, three CG locations are selected:

$$x_{\text{CG}} \in \{x_{\text{NP}} - \Delta, x_{\text{NP}}, x_{\text{NP}} + \Delta\}. \quad (3.4)$$

In the current results, $\Delta = 0.15 \text{ m}$ is used to clearly separate stable and unstable behavior in the simulation. This selection provides one case with $SM > 0$ (stable), one with $SM = 0$ (neutral), and one with $SM < 0$ (unstable).

Chapter 4

Neutral Point Estimation

Under the project assumptions (identical lifting surfaces and negligible downwash interaction), the neutral point (NP) reduces to the midpoint between the wing and tail aerodynamic centers:

$$x_{NP} = \frac{x_{AC,w} + x_{AC,t}}{2}. \quad (4.1)$$

Substituting $x_{AC,w} = 0.0$ m and $x_{AC,t} = 1.76$ m gives:

$$x_{NP} = \frac{0.0 + 1.76}{2} = 0.88 \text{ m}. \quad (4.2)$$

This value is used as the reference CG location for the neutral stability case and as the boundary for stable ($x_{CG} < x_{NP}$) versus unstable ($x_{CG} > x_{NP}$) configurations.

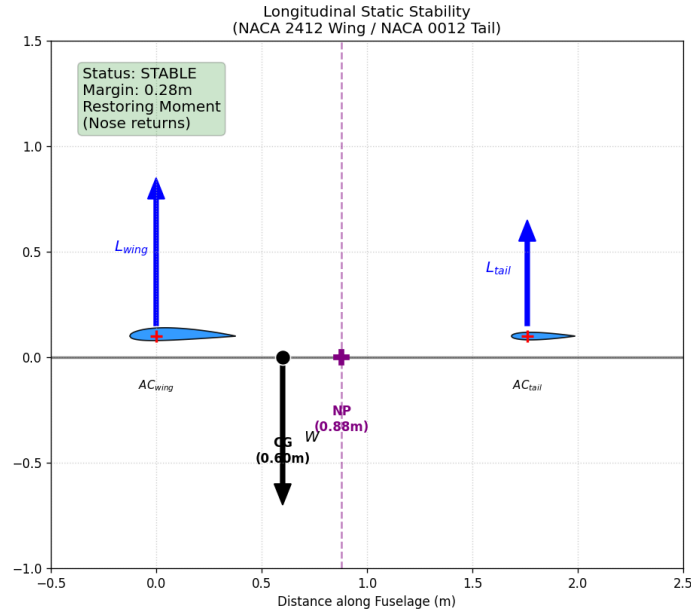


Figure 4.1: Longitudinal stability diagram showing wing AC, tail AC, CG, and the estimated neutral point location.

Chapter 5

Experimental Procedure (Simulation-Based)

This chapter details the numerical procedure used to emulate the required glide-test stability verification. The goal is to reproduce the *stable*, *neutral*, and *unstable* longitudinal behaviors by shifting the center of gravity (CG) relative to the neutral point (NP), while keeping the airframe geometry and aerodynamic assumptions fixed.

5.1 Overview of the Procedure

A nonlinear longitudinal rigid-body model (vertical-plane motion) is simulated by integrating the equations of motion using a Runge–Kutta method (RK45). The CG location x_{CG} is treated as a controlled parameter (representing ballast movement). For each x_{CG} case:

- the model is initialized near a trimmed condition so that the initial pitching moment about the CG is approximately zero, and
- the subsequent pitch response $\theta(t)$ and trajectory are observed to determine whether disturbances decay (stable), remain marginal (neutral), or grow (unstable).

This approach mimics a glide-test conceptually: a configuration that is statically stable should exhibit a restoring tendency after a small disturbance, whereas an unstable configuration should show divergence.

5.2 Equations of Motion

The longitudinal dynamics are expressed in body-axis velocities (u, w) , pitch rate q , and pitch attitude θ . The translational and rotational equations are:

$$m(\dot{u} + qw) = -mg \sin \theta + F_x, \quad (5.1)$$

$$m(\dot{w} - qu) = mg \cos \theta + F_z, \quad (5.2)$$

$$I_{yy}\dot{q} = M, \quad (5.3)$$

$$\dot{\theta} = q, \quad (5.4)$$

where m is the mass, I_{yy} is the pitch-axis moment of inertia, g is gravitational acceleration, F_x and F_z are the aerodynamic force components along the body axes, and M is the aerodynamic pitching moment about the CG. These equations capture the coupling between translational motion and pitch dynamics via the qw and qu terms.

5.3 Aerodynamic Model

The aerodynamic state is primarily determined by the angle of attack α and airspeed magnitude V :

$$\alpha = \arctan\left(\frac{w}{u}\right), \quad V = \sqrt{u^2 + w^2}, \quad q_\infty = \frac{1}{2}\rho V^2, \quad (5.5)$$

where ρ is air density and q_∞ is dynamic pressure.

5.3.1 Lift forces

The wing and tail lifts are modeled using linear lift curves:

$$L_w = q_\infty S_w C_{L\alpha_w} \alpha, \quad L_t = q_\infty S_t C_{L\alpha_t} \alpha_t, \quad (5.6)$$

where S_w, S_t are lifting areas and $C_{L\alpha_w}, C_{L\alpha_t}$ are lift-curve slopes. The tail effective angle α_t includes pitch-rate contribution:

$$\alpha_t = \alpha + \frac{ql_t}{V}, \quad l_t = x_{AC,t} - x_{CG}. \quad (5.7)$$

The term $\frac{ql_t}{V}$ captures the additional local incidence at the tail due to rotational motion, which is important for reproducing realistic pitch response even in a simplified model.

5.3.2 Pitching moment about the CG

The aerodynamic pitching moment is computed from the lift forces acting at their respective aerodynamic centers:

$$M = L_w(x_{CG} - x_{AC,w}) - L_t(x_{AC,t} - x_{CG}). \quad (5.8)$$

This expression directly links stability to geometry: when x_{CG} moves forward or aft relative to the aerodynamic centers, the moment arm signs change, which changes the sign of the static stability derivative $\partial M/\partial\alpha$.

5.4 Trim Initialization

To ensure that the observed response is due to stability properties (and not an initial bias moment), each case is initialized at an approximate trim condition by solving for an angle α_{trim} that makes the net initial moment small:

$$M(\alpha_{\text{trim}}; q = 0, x_{\text{CG}}) \approx 0. \quad (5.9)$$

After computing α_{trim} , the initial velocity components are set as:

$$u_0 = V_0 \cos \alpha_{\text{trim}}, \quad w_0 = V_0 \sin \alpha_{\text{trim}},$$

and the initial attitude is chosen consistently as $\theta_0 = \alpha_{\text{trim}}$ (small flight-path angle assumption). This provides a controlled baseline so that differences between cases arise primarily from CG placement.

5.5 Static Stability Verification Metric

In addition to time-domain observation, static stability is quantified using a numerical approximation of the pitching moment slope with respect to α around the trim point:

$$\frac{\partial M}{\partial \alpha} \approx \frac{M(\alpha_0 + \Delta\alpha) - M(\alpha_0 - \Delta\alpha)}{2\Delta\alpha}. \quad (5.10)$$

This metric directly implements the longitudinal static stability criterion:

$$\frac{\partial M}{\partial \alpha} < 0 \Rightarrow \text{stable}, \quad = 0 \Rightarrow \text{neutral}, \quad > 0 \Rightarrow \text{unstable}.$$

Using this slope provides an objective classification that complements the simulation plots.

5.6 Test Steps

For each CG placement, the following steps are executed:

1. **Select CG position:** choose x_{CG} ahead of NP (stable), at NP (neutral), or aft of NP (unstable).
2. **Trim computation:** solve for α_{trim} such that $M \approx 0$ at $t = 0$ with $q = 0$.
3. **Time integration:** simulate the nonlinear equations using RK45 over a fixed duration.

4. **Response extraction:** record pitch angle $\theta(t)$ and the flight path (position evolution).
5. **Stability metrics:** compute $SM = x_{NP} - x_{CG}$ and evaluate $\partial M/\partial \alpha$ numerically.
6. **Classification:** label each case as stable/neutral/unstable based on the sign of both SM and $\partial M/\partial \alpha$.

This procedure produces (i) time-domain evidence similar to glide-test behavior and (ii) numerical stability indicators that can be compared directly with theoretical predictions.

Chapter 6

Results and Discussion

6.1 Simulation Results

Figure 6.1 presents the simulation outputs for three CG placements relative to the neutral point: a forward CG (stable), CG at NP (neutral), and an aft CG (unstable). Two observables are used as evidence:

- **Flight path (side view):** how altitude evolves with forward distance, which indirectly reflects whether the aircraft is maintaining a reasonable attitude or entering a pitch-divergent trajectory.
- **Pitch angle response:** $\theta(t)$, which is the most direct indicator of longitudinal stability.

In the **stable case** (CG ahead of NP), the pitch angle remains bounded and does not exhibit divergence. The corresponding flight path evolves smoothly without sudden upward/downward excursions caused by excessive pitch-up or pitch-down moments.

In the **neutral case** (CG at NP), the pitch angle shows minimal restoring tendency. A small pitch perturbation is neither strongly corrected nor amplified. This behavior is consistent with marginal static stability, where the aircraft is sensitive to disturbances but does not inherently diverge in the static sense.

In the **unstable case** (CG aft of NP), the pitch angle diverges with time, indicating that the aircraft generates a destabilizing pitching moment in response to perturbations. This pitch divergence directly degrades the flight path, causing the trajectory to depart from a smooth glide-like motion.

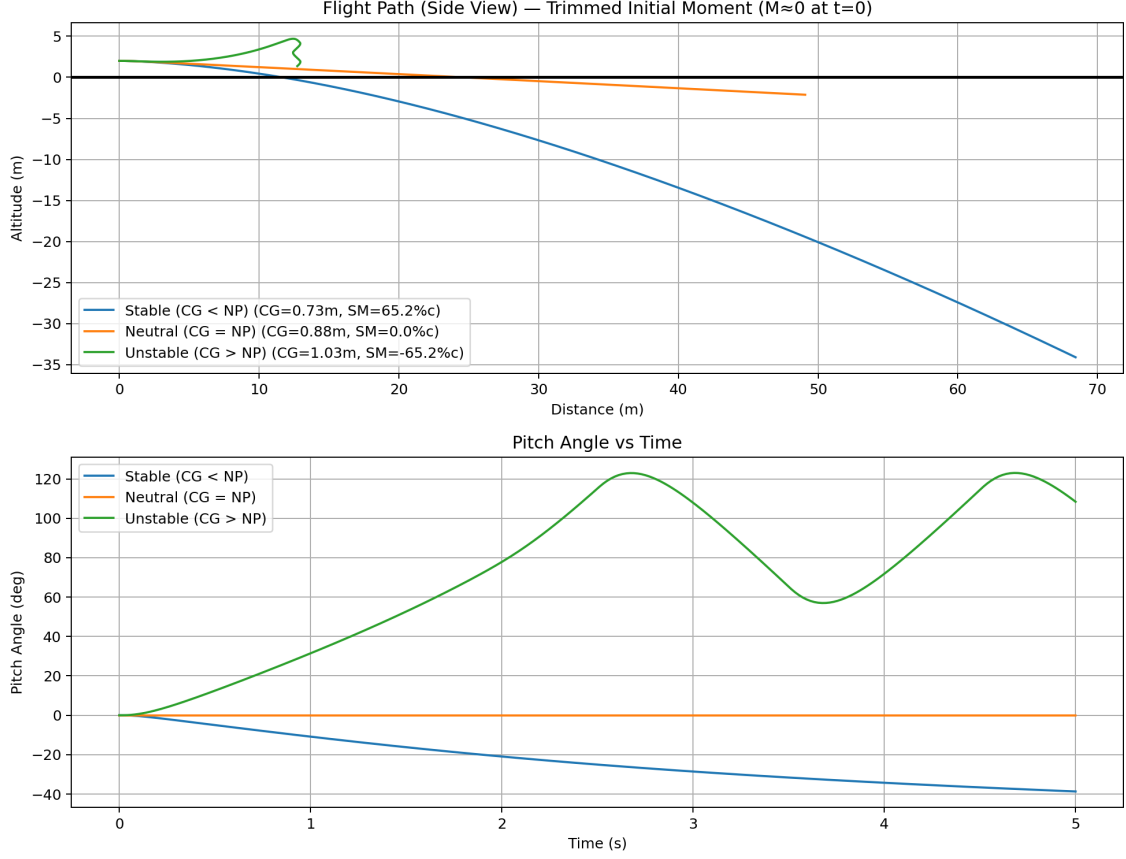


Figure 6.1: Simulation outputs: flight path (top) and pitch angle vs time (bottom) for stable, neutral, and unstable cases.

6.2 Quantitative Stability Summary

To complement the time-domain evidence, stability is quantified using the **static margin** and the numerical estimate of the **static stability derivative** $\partial M / \partial \alpha$. The static margin is defined as $SM = x_{NP} - x_{CG}$, so it is positive for stable CG positions and negative when CG is aft of the NP. The numerical slope $\partial M / \partial \alpha$ is computed around the trim condition using a small perturbation in α and serves as a direct check of the stability criterion: negative slope implies restoring pitching moment, and positive slope implies destabilizing pitching moment.

Table 6.1: Static margin and numerical $\partial M / \partial \alpha$ stability classification.

Case	x_{CG} (m)	x_{NP} (m)	SM (m)	$SM(\%c)$	α_{trim} (deg)	$\partial M / \partial \alpha$ (N·m/rad)
Stable (CG<NP)	0.73	0.88	0.15	65.22	0.0	-31.865
Neutral (CG=NP)	0.88	0.88	0.00	0.00	0.0	0.000
Unstable (CG>NP)	1.03	0.88	-0.15	-65.22	0.0	+31.865

6.3 Discussion

6.3.1 Stable configuration ($x_{CG} < x_{NP}$)

For $x_{CG} = 0.73$ m and $x_{NP} = 0.88$ m, the static margin is $SM = 0.15$ m, indicating that the NP lies behind the CG. In this condition, an increase in angle of attack produces a restoring (nose-down) moment because the effective lift forces act at aerodynamic centers located aft of the CG. This is confirmed by the negative value of $\partial M/\partial \alpha = -31.865$ N·m/rad. The pitch response in Fig. 6.1 remains bounded, showing that disturbances do not grow into divergence. The flight path is correspondingly smooth, consistent with stable gliding behavior.

6.3.2 Neutral configuration ($x_{CG} = x_{NP}$)

For $x_{CG} = x_{NP} = 0.88$ m, the static margin becomes zero. The aerodynamic forces generate nearly zero net pitching-moment slope with respect to α , which is reflected by $\partial M/\partial \alpha \approx 0$. This implies that the aircraft exhibits no inherent static tendency to restore or diverge after a small perturbation. Therefore, the pitch angle response is expected to be sensitive to initial conditions and numerical/physical damping, but it should not show a clear restoring trend. This case serves as a consistency check that the NP computation is correct, because the transition between stable and unstable regimes occurs at this CG location.

6.3.3 Unstable configuration ($x_{CG} > x_{NP}$)

For $x_{CG} = 1.03$ m, the CG is aft of the NP, yielding $SM = -0.15$ m. In this condition, an increase in angle of attack produces a destabilizing (nose-up) tendency because the CG lies behind the effective point where the lift forces act, causing the aerodynamic moment to reinforce the perturbation. This is confirmed by the positive value $\partial M/\partial \alpha = +31.865$ N·m/rad. The pitch angle diverges in Fig. 6.1, and the flight path departs from a steady glide profile as the attitude becomes increasingly unfavorable.

6.3.4 Interpretation of $SM(\%c)$ magnitude

The computed $SM(\%c)$ values are numerically large because the chord is relatively small ($c = 0.23$ m) compared with the chosen CG offsets (± 0.15 m). Since

$$SM(\%c) = \frac{x_{NP} - x_{CG}}{c} \times 100,$$

a fixed dimensional shift produces a large percent-chord margin when c is small. This does not affect the core verification objective of the project, which is the **sign change** of SM and the corresponding sign change of $\partial M/\partial \alpha$ at the NP.

Chapter 7

Comparison with Theoretical Predictions

For the simplified two-surface configuration considered in this project (identical wing and tail lifting characteristics and negligible downwash interaction), the theoretical neutral point (NP) reduces to the midpoint between the aerodynamic centers of the wing and tail:

$$x_{\text{NP}} = \frac{x_{\text{AC},w} + x_{\text{AC},t}}{2} = 0.88 \text{ m.} \quad (7.1)$$

This result is purely geometric under the given assumptions and represents the CG location at which the net pitching-moment slope with respect to angle of attack becomes zero.

7.1 Theoretical Stability Classification

Static stability is determined by the CG position relative to NP:

$$x_{\text{CG}} < x_{\text{NP}} \Rightarrow \text{stable}, \quad x_{\text{CG}} = x_{\text{NP}} \Rightarrow \text{neutral}, \quad x_{\text{CG}} > x_{\text{NP}} \Rightarrow \text{unstable}.$$

This prediction follows directly from the definition of NP as the boundary where $\frac{dC_m}{d\alpha} = 0$ (or equivalently $\frac{\partial M}{\partial \alpha} = 0$). When the CG is ahead of NP, the aerodynamic forces produce a restoring pitching moment after a small increase in α ; when CG is behind NP, the pitching moment reinforces the perturbation.

7.2 Agreement Between Theory and Simulation

The simulation outcomes confirm the theoretical prediction using two independent checks.

1. **Static margin sign (geometric check).** The static margin is defined as

$$SM = x_{\text{NP}} - x_{\text{CG}}.$$

Hence, theory expects $SM > 0$ for stable, $SM = 0$ for neutral, and $SM < 0$ for unstable configurations. The computed margins follow this exactly: the forward

CG case yields positive SM , the NP case yields $SM = 0$, and the aft CG case yields negative SM . This confirms that the chosen CG placements correctly span both sides of the theoretically computed neutral point.

2. **Static stability derivative (aerodynamic moment check).** A second and more direct verification is obtained by evaluating the sign of the numerical static stability derivative $\partial M/\partial\alpha$ around trim. Theory requires:

$$\frac{\partial M}{\partial\alpha} < 0 \Rightarrow \text{stable}, \quad \frac{\partial M}{\partial\alpha} = 0 \Rightarrow \text{neutral}, \quad \frac{\partial M}{\partial\alpha} > 0 \Rightarrow \text{unstable}.$$

The simulation results satisfy this criterion as well: the stable configuration produces a negative slope, the neutral configuration produces a slope approximately equal to zero, and the unstable configuration produces a positive slope. This agreement is significant because it validates the stability transition using the underlying physics (pitching-moment response) rather than geometry alone.

7.3 Conclusion of the Comparison

Because the simulation reproduces both (i) the correct static margin sign change across x_{NP} and (ii) the corresponding sign change of $\partial M/\partial\alpha$, the numerical evidence is consistent with the theoretical prediction that the stability boundary occurs at $x_{\text{CG}} = x_{\text{NP}} = 0.88$ m. Thus, the simulation verifies the expected stable–neutral–unstable transition across the neutral point.

Chapter 8

Conclusions

This project aimed to estimate the neutral point (NP) for a simplified two-surface configuration and to verify the resulting stability transition by shifting the center of gravity (CG) relative to the NP. The following conclusions are drawn.

- **Neutral point estimation:** Under the project assumptions of identical lifting surfaces and negligible downwash interaction, the NP reduces to the midpoint between the aerodynamic centers of the wing and tail. Using the given geometry, the neutral point is obtained as:

$$x_{\text{NP}} = 0.88 \text{ m.}$$

This value represents the theoretical boundary between restoring and diverging static pitching behavior.

- **Stability classification by CG position:** Longitudinal static stability is governed by the relative position of CG and NP. The study confirms the standard static stability classification:

$$x_{\text{CG}} < x_{\text{NP}} \Rightarrow \text{stable}, \quad x_{\text{CG}} = x_{\text{NP}} \Rightarrow \text{neutral}, \quad x_{\text{CG}} > x_{\text{NP}} \Rightarrow \text{unstable}.$$

In physical terms, when the CG is forward of the NP, a small increase in angle of attack generates a restoring pitching moment; when CG is aft of NP, the moment reinforces the perturbation.

- **Verification using dynamic response and static derivatives:** The stability transition was verified using two complementary forms of evidence obtained from the numerical simulation:
 - (i) the time-domain response (bounded pitch angle and smooth flight path for the stable case, marginal behavior near NP, and pitch divergence for the unstable case), and
 - (ii) the numerical static stability derivative evaluated around trim, where the sign of $\partial M / \partial \alpha$ was negative, approximately zero, and positive for the stable, neutral, and unstable cases respectively.

The agreement of both checks indicates that the simulation reproduces the theoretical stability boundary at $x_{\text{CG}} = x_{\text{NP}}$.

Deliverable note: The project specification requests physical glide-test photos/videos as experimental evidence. Due to practical constraints, this submission provides simulation-based evidence using the same geometry and mass assumptions. The results still demonstrate the required stable and unstable regimes clearly and verify the neutral point prediction through consistent static and dynamic indicators.

Appendix A

Experimental Evidence (Simulation Evidence)

The deliverable requests photos of stable and unstable glide tests. In this submission, the evidence is provided in the form of simulation outputs that replicate the glide-test behavior using the same geometry and mass assumptions.

A.1 Stability Diagram Evidence

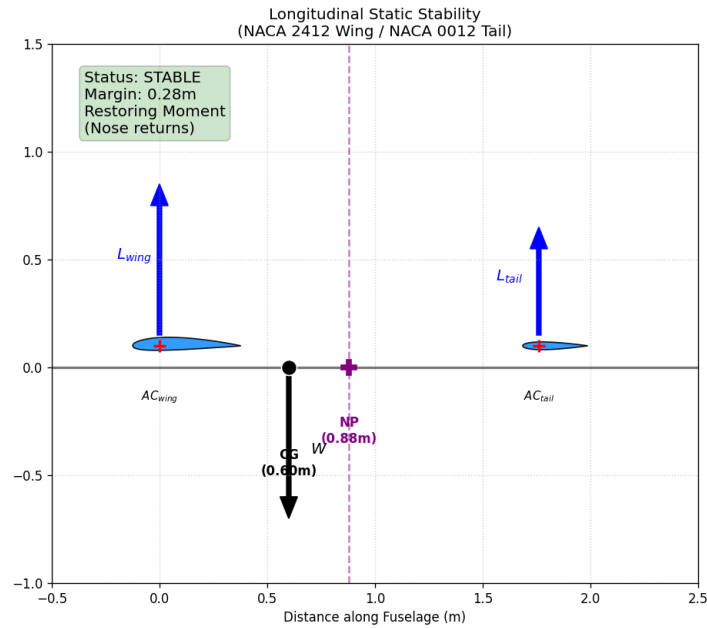


Figure A.1: Diagram showing wing AC, tail AC, CG, and NP, including restoring/diverging moment interpretation.

A.2 Stable vs Unstable Evidence

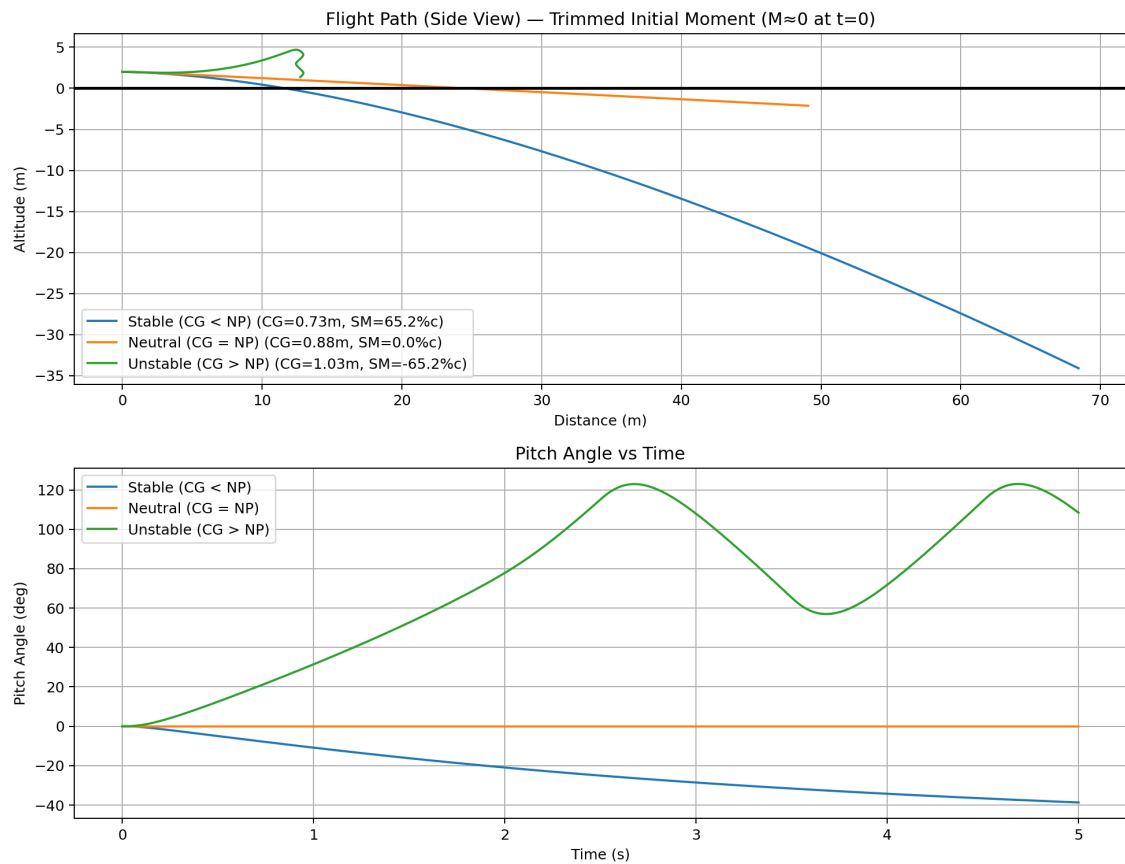


Figure A.2: Evidence plots showing stable, neutral, and unstable longitudinal responses.